

Experiment: Simple Linear Regression

Experiment Title

Implementation and Evaluation of Simple Linear Regression

Objective

The objective of this experiment is to implement a Simple Linear Regression model to predict a continuous target variable, visualize the best-fit regression line, and evaluate the model using standard regression performance metrics.

Problem Statement

House prices generally increase with an increase in the size of the house. In this experiment, you will develop a Simple Linear Regression model to predict the price of a house based on its size.

Use the following dataset to build, train, and evaluate the regression model.

Table 1: House Price Dataset

House Size (sq.ft.)	House Price (Lakhs)
600	25
800	32
1000	38
1200	45
1400	52
1600	58
1800	65
2000	72
2200	80
2400	87

Tasks

1. Create the dataset using Pandas and display the first few records.
2. Perform exploratory data visualization by plotting a scatter plot between House Size and House Price. Briefly comment on the relationship between the variables.
3. Split the dataset into training and testing sets using an 80:20 ratio.
4. Build a Simple Linear Regression model using the training data.
5. Display the learned model parameters:
 - Regression Constant (θ_0)

- Regression Coefficient (θ_1)
6. Predict the house prices for the test dataset and compare the predicted values with the actual values.
 7. Predict the house price for the following house sizes:
 - 1500 sq.ft.
 - 1700 sq.ft.
 - 2500 sq.ft.
 8. Plot the regression line along with the original data points and interpret the obtained graph.
 9. Evaluate the performance of the model using the following metrics:
 - Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - Coefficient of Determination (R^2 Score)
 10. Plot the Actual vs. Predicted values and briefly comment on the model's prediction accuracy.